

Unit 5: Square Root Functions

Study Guide – Concepts & Examples

Name: _____

Date: _____

Period: _____

✓ You Need to Be Able To...

- | | |
|---|---|
| ☐ Simplify radical expressions (square roots and cube roots) | ☐ Find the inverse of a square root function |
| ☐ Convert between radical form and rational exponent form | ☐ Solve square root equations algebraically |
| ☐ Identify the domain and range of a square root function | ☐ Identify extraneous solutions and check for validity |
| ☐ Identify transformations of square root functions | ☐ Write the equation of a square root function given its end-point and a point |
| ☐ Identify the endpoint of a square root function | |
| ☐ Use composition to determine if two functions are inverses | |

1. Simplifying Radicals

Key Concept: Find perfect square (or perfect cube) factors and pull them out.

- $\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$
- $\sqrt{x^2} = |x|$ (or x if $x \geq 0$)
- $\sqrt[3]{x^3} = x$

Examples:

$$\begin{aligned}\sqrt{72} &= \sqrt{36 \cdot 2} = 6\sqrt{2} \\ \sqrt{50x^4} &= \sqrt{25 \cdot 2 \cdot x^4} = 5x^2\sqrt{2} \\ \sqrt[3]{54x^5} &= \sqrt[3]{27 \cdot 2 \cdot x^3 \cdot x^2} = 3x\sqrt[3]{2x^2} \\ \sqrt{48a^3b^4} &= 4ab^2\sqrt{3a}\end{aligned}$$

2. Radical $\leftrightarrow$ Rational Exponents

Key Concept:

$$\boxed{\sqrt[n]{x^m} = x^{m/n} \quad \text{and} \quad x^{m/n} = \sqrt[n]{x^m}}$$

The **denominator** = index (root)

The **numerator** = power

Examples:

$$\begin{aligned}\sqrt{x} &= x^{1/2} & x^{3/4} &= \sqrt[4]{x^3} \\ \sqrt[3]{x^5} &= x^{5/3} & (\sqrt{x})^3 &= x^{3/2}\end{aligned}$$

3. Domain and Range

Key Concept: For $f(x) = a\sqrt{x-h} + k$:

- **Domain:** Set radicand ≥ 0 . Domain starts at $x = h$.
- **Range:**
 - If $a > 0$: $y \geq k$
 - If $a < 0$: $y \leq k$

Examples:

$$\begin{aligned}f(x) &= \sqrt{x+6} - 5 \\ \Rightarrow D : x &\geq -6, \quad R : y \geq -5 \\ f(x) &= -2\sqrt{x-3} + 4 \\ \Rightarrow D : x &\geq 3, \quad R : y \leq 4 \\ f(x) &= \frac{1}{3}\sqrt{x+2} - 1 \\ \Rightarrow D : x &\geq -2, \quad R : y \geq -1\end{aligned}$$

4. Transformations

Key Concept: For $f(x) = a\sqrt{x-h} + k$ vs. $y = \sqrt{x}$:

- $h > 0$: Right h • $h < 0$: Left $|h|$
- $k > 0$: Up k • $k < 0$: Down $|k|$
- $|a| > 1$: Vertical stretch
- $0 < |a| < 1$: Vertical compression
- $a < 0$: Reflect over x -axis

Examples:

$$\begin{aligned}f(x) &= \sqrt{x-3} + 2 \\ \Rightarrow &\text{Right } 3, \text{ Up } 2 \\ f(x) &= -4\sqrt{x+8} - 5 \\ \Rightarrow &\text{Left } 8, \text{ Down } 5, \text{ Reflect,} \\ &\text{Vertical stretch by } 4 \\ f(x) &= \frac{1}{2}\sqrt{x-1} + 7 \\ \Rightarrow &\text{Right } 1, \text{ Up } 7, \\ &\text{Vert. compression by } \frac{1}{2}\end{aligned}$$

5. Identifying Endpoints

Key Concept: The endpoint of $f(x) = a\sqrt{x-h} + k$ is:

$$(h, k)$$

This is where the domain begins and the graph starts.

Examples:

$$f(x) = \sqrt{x-2} + 4 \Rightarrow (2, 4)$$

$$f(x) = -3\sqrt{x+5} - 1 \Rightarrow (-5, -1)$$

$$f(x) = \sqrt{x} + 6 \Rightarrow (0, 6)$$

6. Composition to Check Inverses

Key Concept: f and g are inverses if and only if:

$$f(g(x)) = x \text{ AND } g(f(x)) = x$$

Both must equal x .

Example: Are $f(x) = x^2 + 2$ and $g(x) = \sqrt{x-2}$ inverses?

$$f(g(x)) = (\sqrt{x-2})^2 + 2 = x \checkmark$$

$$g(f(x)) = \sqrt{x^2} = |x| \times$$

NOT inverses (only works for $x \geq 0$).

7. Finding the Inverse

Steps:

1. Replace $f(x)$ with y
2. Switch x and y
3. Isolate the radical, then square both sides
4. Replace y with $f^{-1}(x)$

Example: Find the inverse of $f(x) = \sqrt{x+2} - 4$

$$y = \sqrt{x+2} - 4$$

$$x = \sqrt{y+2} - 4 \quad (\text{switch})$$

$$x + 4 = \sqrt{y+2} \quad (\text{isolate})$$

$$(x+4)^2 = y+2 \quad (\text{square})$$

$$f^{-1}(x) = x^2 + 8x + 14$$

8. Solving Square Root Equations

Steps:

1. Isolate the radical
2. Square both sides
3. Solve the resulting equation
4. **CHECK all solutions! (REQUIRED)**

Example 1: $\sqrt{2x+7} + 3 = 8$

$$\sqrt{2x+7} = 5$$

$$2x+7 = 25$$

$$x = 9$$

Check: $\sqrt{25} + 3 = 8 \checkmark$

Example 2 (Extraneous): $\sqrt{-x-4} - 4 = x$

$$\sqrt{-x-4} = x+4$$

$$-x-4 = (x+4)^2$$

$$0 = x^2 + 9x + 20 = (x+4)(x+5)$$

$$x = -4 \text{ or } x = -5$$

Check $x = -4$: $\sqrt{0} - 4 = -4 \checkmark$

Check $x = -5$: $\sqrt{1} - 4 = -3 \neq -5 \times$

Solution: $x = -4$ only

Example 3 (Rational Exp.): $8 = 3(x-2)^{1/2} - 4$

$$12 = 3(x-2)^{1/2}$$

$$4 = (x-2)^{1/2}$$

$$16 = x-2$$

$$x = 18$$

9. Writing Equations from Info

Given: Endpoint (h, k) and another point.

Use: $f(x) = a\sqrt{x-h} + k$

Steps:

1. Endpoint gives h and k
2. Substitute the other point for x and $f(x)$
3. Solve for a

Example: Endpoint $(2, 3)$, passes through $(6, 1)$.

$$f(x) = a\sqrt{x-2} + 3$$

$$1 = a\sqrt{6-2} + 3$$

$$1 = 2a + 3$$

$$a = -1$$

$$f(x) = -\sqrt{x-2} + 3$$